MATEMAFIJA

Final Five-Minute Speaking Script

3 minutes story + 2 minutes prototype demo

Galina and Gleb — VILNIUS TECH

Hack4Vilnius 2026 — final repository state

Target: calm delivery, approximately 650 spoken words including demo narration. Cues in brackets are not spoken. Galina owns the story and city/business value; Gleb owns the live evidence and mechanism demo. MCU, contributions and response effects are simulated unless explicitly labelled observed.

Slide 1 — Team and hook

GALINA | 0:00–0:18

Imagine it is a weekday evening in Vilnius. Your car needs charging before tomorrow, but your usual station has no available charging point. Do you wait, keep searching, or drive somewhere else? We are Galina and Gleb, Applied Mathematics students from VILNIUS TECH, and this is MATEMAFIJA.

Slide 2 — The city problem

GALINA | 0:18–0:52

Vilnius already has charging infrastructure and public data, but availability is uneven. Our current collection contains more than 2.8 million OCPI status observations and over 2,100 current EVSE records at locations marked Vilnius. Yet the data do not show a physical queue, individual waiting time or why a driver leaves. So simply inventing a congestion fee would be easy — and scientifically weak.

Slide 3 — The solution

GALINA | 0:52–1:35

We built a policy stress-testing digital twin. It turns live charging data into three simple steps: detect sustained scarcity, create a small and capped contribution, and use funded credits to offer drivers a less-crowded alternative. Instead of only saying “pay more”, MATEMAFIJA asks: can a useful alternative for the driver also relieve pressure on the network?

Slide 4 — Why it is credible and useful

GALINA | 1:35–2:25

The key is transparency. We know station status, time and tariff from public data. We do not yet know how strongly people will react to a price or a credit, so we systematically vary possible response strengths

instead of hiding the uncertainty. A scenario is promising only if availability improves without moving scarcity elsewhere, while alternatives remain practical and every credit remains funded.

The daily beneficiaries are drivers. Charging operators and the city are the first potential customers of the policy-testing and pilot-design tool. Our headline pilot KPI is simple: reduce qualified zero-availability without increasing scarcity at receiver stations.

Slide 5 — Demo: observed evidence

GLEB | 2:25–3:18

[Open Evidence. Show the map, source timestamp and readiness panel.] Here is the live evidence layer. These are source records, not a mock dataset. We show the map, current states and real tariff structure, and we keep equipment faults separate from demand-related scarcity. Our collection currently spans just over four days; the formal model requires seven. The formal effectiveness result is therefore not yet available. Instead, the next screen shows an illustrative preview, clearly separated from evidence.

Slide 6 — Demo: policy, response and credit flow

GLEB | 3:18–4:25

[Open Policy, then Response. Move one credit or response control only.] When scarcity persists, OSA-v1 creates a bounded signal after a fifteen-minute grace period. It is not an actual bill. The Policy view shows how contributions can fund credits for available alternatives. In Response, we vary how likely a driver is to release a busy point or choose another station. These are visible assumptions, not measured Vilnius behaviour.

[Open Robustness. Show the labelled preview and move one response value.] This heatmap answers one honest question: how strong would the response need to be before the mechanism becomes interesting? The preview is based on declared assumptions, not observed effectiveness or a forecast. The production model then checks whether scarcity falls without simply moving it elsewhere, and whether every simulated credit remains funded.

Slide 7 — Benefit, business path and ask

GALINA | 4:25–5:00

MATEMAFIJA gives Vilnius a practical decision: which sites should enter a small city–operator pilot, what response must be measured, and when the policy is not worth scaling. Session data and privacy-safe choice records would turn today's assumptions into measured Vilnius behaviour.

Lithuania's DAEI system shows that verified transport contributions can be accounted for, and it is a possible future bridge. Our credits are separate today. We are not asking the city to trust a formula. We are giving it a safe way to test one. MATEMAFIJA — turning scarcity into better charging choices. Thank you.

Seven-slide production plan

	Slide	Visual proof
1	Team + hook	Vilnius evening, “No available point — what now?”
2	Problem	Marta/origin situation plus one live-data number
3	Solution	Observe $\rightarrow$ qualify scarcity $\rightarrow$ fund MCU $\rightarrow$ offer alternative
4	Credibility	observed/reconstructed/assumed/simulated and four gates
5	Evidence demo	real map, source timestamp, tariffs and readiness boundary
6	Mechanism demo	OSA signal, one response control and labelled illustrative preview
7	Benefit + ask	Drivers / CPO / Vilnius, one KPI, bounded pilot, DAEI future path

Recording rules

- Rehearse the full clicks and speech at least three times; target 4:45–4:55.
- Keep only one live control movement. Preload the database; never wait for network collection.
- Record a clean demo clip and save screenshots of Evidence, Policy, Response and Decision.
- Update the rounded observation count only if it remains easy to say and verify.
- Never call zero availability a queue; never call MCU money, cashback or DAEI.
- Submit the video as Public or Unlisted and verify the link in an incognito window.

Fast jury answers

What is innovative? We test a complete, budget-constrained scarcity-to-credit circulation mechanism, including behavioural uncertainty, receiver spillover and accounting — not only a fee formula.

Who uses and pays? Drivers benefit; a CPO and the city use the policy lab. The first commercial hypothesis is a city–operator pilot and decision-support engagement. Real funding and liability must be contracted before credits carry value.

Can you prove behaviour changes? No. We expose assumed response parameters and define the minimum response that a pilot must validate.

Why show a heatmap before seven days? The formal effectiveness result remains locked. The visible chart is an illustrative response surface generated from declared assumptions, permanently labelled as not evidence and not a forecast.

Are credits real? No. MCU is an integer simulation unit with reconciled lifecycle states. The prototype processes no payment and holds no personal wallet.

Is this DAEI? No. DAEI is a regulated renewable-transport accounting system. It is a precedent and possible future integration path, not the present MCU mechanism.

What do you need next? Qualified pilot sites, an operator partner, Session/CDR evidence and privacy-safe choice/redemption measurements.

Final verbal claim: Under declared response assumptions, MATEMAFIJA identifies when a contribution-to-credit policy is jointly robust within observable system, redistribution, user-proxy and accounting limits, and turns the uncertainty into a measurable Vilnius pilot.